

Practical No. 2

Aim:- To understand & implement Views & Indexing in MySQL for performance optimization in a student Course Management system, enabling faster queries & simplified reporting.

Theory :-

① Views :-

- A view is a virtual table based on the result of a SELECT query.
- Advantages:- Simplifies complex queries, enhances security, allows limited updates/deletes.
- Update/Delete is allowed only if the view is based on a single table without aggregation.

② Indexing :-

- An index improves search & query performance.
- Types:- primary key, Unique, Normal, Composite.
- Advantages:- faster SELECT query better JOIN performance.
- Trade-off:- slightly slower INSERT/UPDATE/DELETE; extra storage usage.

③ Optimization :-

- Views make reporting easy without repeatedly writing complex queries.
- Indexes reduce query execution time on large tables.
- Combined, they enhance database efficiency.

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Conclusion

- Views simplify reporting Δ allow limited updates/deletes on virtual tables.
- Indexes significantly improve query performance, especially on frequently searched columns.
- Combining views Δ indexes optimizes the student Course Management System for faster data retrieval.
- Lab successfully demonstrates performance improvement Δ easier data management using views & indexes.